**1** The table shows information about the number of goals a team scored in 38 games.

| Points | Frequency |
|---|---|
| 0 | 8 |
| 1 | 16 |
| 2 | 9 |
| 3 | 5 |
| 4 or more | 0 |

(a) Find the median number of goals scored.

1

(1)

(b) Write down the mode

1

(1)

(c) Work out the total number of goals the team scored in all 38 games.

$0 \times 8 = 0$

$1 \times 16 = 16$

$2 \times 9 = 18$

$3 \times 5 = 15$

1 for points x frequency

$16 + 18 + 15 = 49$

49

(2)

**(Total for Question 1 is 4 marks)**

**2** A biased spinner can land on 1, 2, 3 or 4.
The table shows the probabilities that the spinner will land on 2 and 4.

| **Number** | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Probability** | 0.32 | 0.34 | 0.16 | 0.18 |
| | $2x$ | | $x$ | |

The probability that the spinner will land on 1 is **twice** the probability that the spinner will land on 3.

(a) Complete the table.

$0.34 + 0.18 = 0.52$

$1 - 0.52 = 0.48$

$3x = 0.48$

$x = 0.16$

$2 \times 0.16 = 0.32$

(2)

1 for 0.16

Johnny is going to spin the spinner 200 times.

(b) Work out an estimate for the number of times the spinner will land on 2.

$0.34 \times 200 = 68$

68

1 for 0.34x200

(2)

**(Total for Question 2 is 4 marks)**

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**3** Solve $3(2x + 1) > 27$

$6x + 3 > 27$

$-3 \quad -3$

$6x > 24$

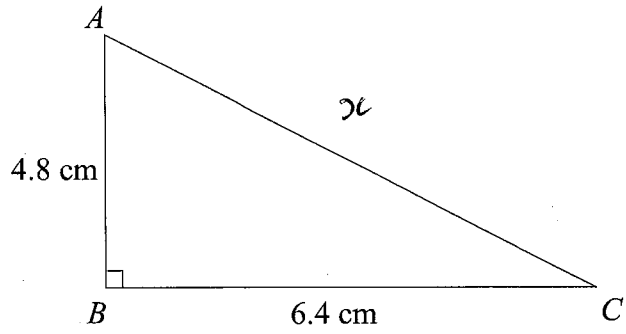
$x > 4$

1 for 4

$x > 4$

**(Total for Question 3 is 2 marks)**

**4**




Calculate the length of $AC$.

$4.8^2 + 6.4^2 = x^2$ 1 for correct substitution

$64 = x^2$ 1 for 64

$x = \sqrt{64}$

$= 8$

8 cm

**(Total for Question 4 is 3 marks)**

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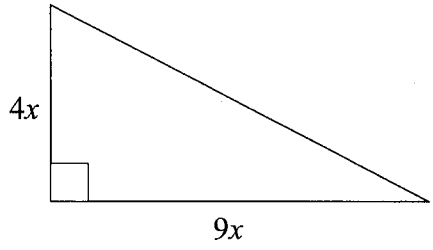
**5** The diagram shows a right angled triangle.




The area of the triangle is 252 cm$^2$

Work out the value of $x$.

$\frac{1}{2} \times 4x \times 9x = 252$

1 for correct area formula $18x^2 = 252$

1 for $x^2 = 14$ $x^2 = 14$

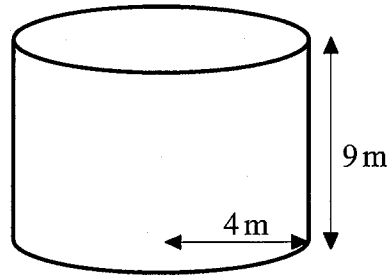
$x = 3.74$ cm

$x = 3.74$ cm

**(Total for Question 5 is 3 marks)**

**6** A solid cylinder has a radius of 4 m and a height of 9 m.

Work out the total surface area of the cylinder.
Give your answer in terms of $\pi$.




Total surface area $= 2\pi r^2 + 2\pi r h$

$= 2\pi(4)^2 + 2\pi(4)(9)$

$= 104\pi \text{ m}^2$

1 for correct area of circle = 50.265 or $\pi(4)^2$

AND correct area of rectangle = 226.19 or $2\pi(4)(9)$

1 for $m^2$

$104\pi \text{ m}^2$

**(Total for Question 6 is 3 marks)**

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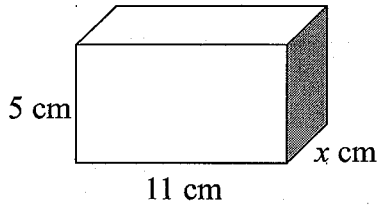
**7** The diagram shows a cuboid.




The volume of the cuboid is 330 $cm^3$

1 for correct formula

Calculate the value of $x$

$5 \times 11 \times x = 330$

$55x = 330$

$x = 6$

6

**(Total for Question 7 is 2 marks)**

**8** Here are the first 5 terms of a sequence.

4 11 18 25 32

Find an expression, in terms of $n$, for the $n$th term of this sequence.

7n 7 14 21 28 35

1 for 7n

$7n - 3$

**(Total for Question 8 is 2 marks)**

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**9** Fearne invests £4500 in a savings account.
She gets 2.5% per annum compound interest.

After $\boldsymbol{n}$ years, Fearne has £5482.81 in her account.
Work out the value of $\boldsymbol{n}$.

1 for any correct formula

$4500 \times 1.025^n = 5482.81$

$\cancel{1.025^n = 1.2184...}$

$4500 \times 1.025^5 = 5091.34$

$4500 \times 1.025^6 = 5218.62$

$4500 \times 1.025^7 = 5349.09$

$4500 \times 1.025^8 = 5482.81$ ✓

8

**(Total for Question 9 is 2 marks)**